Boundary-Aware Spatial and Frequency Dual-Domain Transformer for Remote Sensing Urban Images Segmentation

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***Abstract*— Semantic segmentation of remote sensing (RS) images refers to labeling each pixel with a class to identify objects or land cover types. Existing mainstream spatial-domain semantic segmentation methods are mainly categorized into convolutional neural network (CNN)-based and vision transformer (ViT)-based approaches. The former excels at capturing local features, while the latter is adept at extracting global features. Several recent approaches consider combining CNN and ViT to efficiently capture local and global features. However, these approaches still struggle to capture complete features of the RS images, resulting in inaccurate segmentation. To address this issue, we introduce the fast Fourier transform (FFT), which transforms images into the frequency domain for feature extraction, acquiring the image-size receptive field that can complement spatial-domain methods. Based on this, we propose a boundary-aware spatial and frequency dual-domain transformer, termed dual-domain transformer. Specifically, our dual-domain transformer incorporates a dual-domain mixer (DualM), where the spatial-domain branch combines depthwise convolution and the attention mechanism to extract local and global features effectively, while the frequency-domain branch uses FFT to extract image-size features. The two branches complement each other, enabling a more comprehensive feature extraction of RS images. Meanwhile, a boundary-guided training strategy utilizing a boundary-aware module (BAM) is devised to constrain the model extract and predict boundary detail texture, which is an auxiliary task. In addition, the decoder incorporates a scale-feature fusion module (SFM) for adaptive information fusion between the encoder and decoder. Comprehensive experiments on the Zeebrugge and ISPRS datasets, including Vaihingen and Potsdam, showcase that the dual-domain transformer significantly outperforms state-of-the-art (SOTA) methods.**

***Index Terms*— Fast Fourier transform (FFT), frequency domain representation, remote sensing (RS) image, semantic segmentation.**

I. Introduction

RECENT advancements in unmanned aerial vehicle (UAV) aerial photography have significantly enhanced the quality of remote sensing (RS) images, allowing them to contain richer semantic information and spatial details [1], [2], [3]. Semantic segmentation of RS images, a vital application in this field, refers to classifying each pixel in the image into specific categories based on its semantic meaning. This process facilitates in-depth geographical analysis and supports applications like land cover mapping [4], urban planning [5], and environmental monitoring [6]. Traditional methods [7], [8], [9] to solve the above problems rely on hand-crafted features, texture analysis, and model-based techniques. However, these approaches have limitations in feature representation, require significant manual intervention, and struggle to capture complex spatial-contextual relationships.

To address these challenges, deep learning techniques have been employed for RS image segmentation, predominantly falling into two categories: convolutional neural network (CNN)-based approaches and vision transformer (ViT)-based strategies. CNN-based methods [12], [13], [14], [15], [16], [17] employ multilayer convolutions to process RS images, excelling in capturing spatial hierarchies and details. However, they are limited by local receptive fields. In contrast, ViT-based approaches [18], [19], [20] leverage attention mechanisms for global context understanding, yet they split the image into a patch sequence, destroying the structural features and local information retention. Thus, it is crucial to develop a practical semantic segmentation approach for RS images that simultaneously preserves local features and accommodates global receptive fields. This need gives rise to hybrid CNN-ViT methods, combining CNN for local information extraction and ViT for global context understanding. Wang et al. [11] introduced UnetFormer, enhancing feature representation via novel

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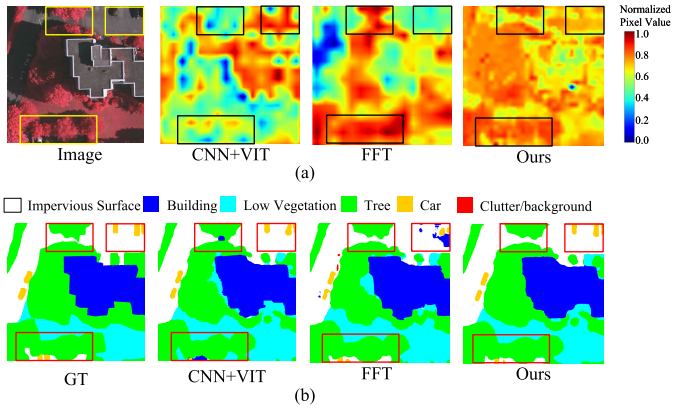


Fig. 1. (a) Visualization of the feature map in the encoder's final layer. Colors represent normalized pixel values, with red regions indicating high network attention. From left to right, it shows the original image, the feature map from the spatial-domain method (hybrid CNN-ViT [10]), the feature map from our implementation of the FFT following the architecture outlined in [11]), and the feature map from our method. (b) GT and the segmentation results corresponding to the different methods in (a). The analysis illustrates that the spatial-domain and frequency-domain methods are complementary for feature extraction, with segmentation more accurate in areas receiving high attention. Our method leverages both domains to jointly extract features, capturing more comprehensive features.

global-local attention. Other methods, such as STUNet [21], CG-Swin [22], and DCSwin [10], integrate hierarchical Swin transformer [23] as a backbone network, coupled with CNN for comprehensive understanding. To summarize, all these methods operate in the spatial domain, analyzing pixel-level information directly from images for semantic segmentation.

However, existing spatial-domain methods still face challenges due to objects with high similarity and some complex scenes in RS images. These challenges arise because spatial-domain methods capture incomplete image features, resulting in imprecise segmentation. As illustrated in Fig. 1(a), the hybrid CNN-ViT approach [10] fails to accurately capture features in shadowed regions, such as the "trees" and "cars," resulting in pixel misclassification and an inability to identify the "cars" obscured by shadows. To address this limitation, we introduce FFT, which transforms images into the frequency domain, enabling comprehensive feature extraction. The FFT can capture various frequency components in the frequency domain [24], [25], [26], [27], [28], obtaining an image-size receptive field. On the one hand, FFT demonstrates excellent segmentation performance in capturing local edge details due to its exceptional ability to extract high-frequency information. As illustrated in Fig. 1, the FFT more accurately segments the edges of building roofs. On the other hand, the FFT can effectively capture features overlooked by the hybrid CNN-ViT method. For example, it accurately identifies the shaded area of the "tree" [see Fig. 1(a)], resulting in more precise segmentation [see Fig. 1(b)]. Therefore, we consider integrating spatial-domain techniques with frequency-domain methods to achieve more comprehensive feature extraction from RS images.

Based on the aforementioned analysis, we propose a boundary-aware spatial and frequency dual-domain transformer (dual-domain transformer) for RS semantic segmentation. Specifically, our dual-domain transformer encoder adheres to the ViT structure but substitutes the traditional self-attention mechanism with a spatial–frequency dual-domain mixer (DualM) mechanism. The spatial-domain branch combines convolution and the attention mechanism to effectively extract local and global features, while the frequency branch uses FFT to extract features of the image-size receptive field. The two branches complement each other, ensuring comprehensive feature extraction. As visualized in Fig. 1, our method captures more comprehensive features and achieves more accurate segmentation. In addition, to enhance edge localization and segmentation, we introduce a boundary-guided training strategy. Utilizing the boundary-aware module (BAM), our dual-domain transformer uses the boundary detail texture to guide training as an auxiliary task. Finally, we introduce a scale-feature fusion module (SFM) in the CNN-based decoder to adaptive merge detailed information from the encoder with its deep semantic information. With all these strategies, our dual-domain transformer attains superior semantic segmentation efficacy in RS imagery, underscoring its robust application potential.

In summary, this article illustrates insights and contributions that are as follows.

1) We introduce FFT and develop a spatial–frequency dual-domain transformer for comprehensive feature extraction in the semantic segmentation of RS images. Extensive experiments validate its effectiveness, surpassing state-of-the-art (SOTA) methods.
2) We propose a boundary-guided training strategy that utilizes a BAM to extract multiscale edge details, enabling the model to better distinguish between neighboring objects and ambiguous boundaries.
3) In the CNN-based decoder, we design the SFM to adaptively merge detailed information from the encoder with its deep semantic information.

II. Related Work

A. Semantic Segmentation of RS Imagery

Semantic segmentation is an essential application of RS image processing, which focuses on pixel-level classification of RS images. Traditional methods in RS often rely on manually crafted features, texture analysis, and model-based techniques with classifiers. However, these approaches demand expert knowledge and substantial time, rendering them inefficient for feature extraction. Recently, deep learning methods have revolutionized this field, automating precise pixel-level classifications and outperforming traditional methods even in challenging environments. These deep learning methods for semantic segmentation primarily fall into two categories: CNN-based and ViT-based models.

CNN-based methods [12], [13], [14], [15], [16], [17] excel at efficiently capturing hierarchical features and local information, which proves beneficial in handling RS images featuring multiscale object characteristics. Nonetheless, their limited receptive fields primarily focus on local details, potentially hindering the capture of crucial long-range dependencies. To address this limitation, researchers have integrated attention mechanisms into CNN-based models, enhancing capturing global context by selectively emphasizing key image regions.

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For instance, AFNet [29] employs attention to augment multiscale feature fusion and layer information integration, thereby improving RS image segmentation. Motivated by the fusion concept, Yang et al. [30] employed attention mechanisms to fuse high-level and low-level semantics, while SAPNet [31] utilizes joint attention to simultaneously model spatial and channel information. Although CNN-based attention methods refine performance, they introduce computational overhead due to model complexity. To mitigate this, MAResU-Net [32] and MSCSANet [33] incorporate linear attention, reducing complexity while maintaining effectiveness. However, due to the over-dependence on convolutional operations, the attention mechanism limits the global feature representations, resulting in the inability to model comprehensive global information effectively.

In recent years, ViT [34] has brought rapid development to RS image segmentation due to its powerful global modeling capability. Leveraging attention mechanisms, ViT-based methods [18], [19], [20] excel at capturing long-range dependencies, which is essential for understanding intricate spatial relationships. However, they segment the image into a sequence of patches, compromising structural features and local information retention. Currently, most ViT-based semantic segmentation methods for RS images follow the classical encoder–decoder structure and have made incredible breakthroughs by combining with CNNs for comprehensive local and global information extraction. Notably, Zhang et al. [35] pioneered a hierarchical hybrid network combining CNN and ViT. Subsequently, to reduce the computational complexity of ViT, UnetFormer [11] introduces approximate computation methods, while DSCTs [36] employ knowledge distillation [37] to combine CNN and transformer. To further obtain hierarchical global information, Swin transformer integrates a window focusing mechanism into RS image segmentation tasks, yielding notable outcomes when combined with CNN, as demonstrated by STUNet [21], CG-Swin [22], and DCSwin [10]. Along these lines, EGUNet [38], DWin-HRFormer [39], and CGVT [40] design more efficient parallel multibranch, multidirectional strip-window attention to reduce model complexity. While these methods focus solely on feature extraction in the spatial domain, they often fail to capture comprehensive features. In contrast, our method can analyze RS images in both frequency and spatial domains, enabling a more comprehensive feature extraction of RS imagery.

B. Deep Learning in the Frequency Analysis

The frequency analysis method initially applied in digital signal processing [41], [42], holds fundamental significance in classical control theory. Recently, frequency methods have been integrated into vision tasks to accelerate the training speed [43], [44], improve model generalization [45], [46], [47], and enhance understanding of nonlocal information [26], [48]. Differing from spatial-domain techniques, frequency-domain analysis effectively separates different frequency components, enabling the extraction of image features overlooked in the spatial domain, which has demonstrated promising outcomes in various vision tasks. For example, Chi et al. [25] proposed the fast Fourier convolution (FFC) by employing the 2-D FFT to capture nonlocal receptive fields, significantly enhancing image classification and video action recognition. LaMa [26] extended FFC for large mask inpainting, improving receptive field coverage and generalization. Notably, HFDT [24] and Fourmer [28] utilize FFT to design a hierarchical transformer for image restoration, while SFMNet [27] employs Fourier-based method to enhance global modeling in face super-resolution.

Recently, researchers have increasingly explored frequency domain analysis to uncover deeper semantic associations. Some notable endeavors have applied frequency analysis directly to spatial-domain methods, yielding remarkable outcomes. For instance, HiLo [49] effectively decouples low and high-frequency information, employing a two-way attention mechanism to capture frequency information. Similarly, AFFormer [50] designs low-pass and high-pass filters based on spectral correlation theory and utilizes a frequency domain correlation kernel to adaptively extract frequency information directly in the spatial domain. Inspired by the successful application of frequency analysis in various domains, we endeavor to apply FFT frequency domain methods to RS image segmentation, combining frequency and spatial domains for comprehensive feature extraction.

C. Boundary Enhancement in Semantic Segmentation of RS Images

Boundary enhancement is crucial for semantic segmentation in RS images, as it significantly enhances the delineation between distinct regions, thus improving pixel classification accuracy [51]. Initially, researchers predominantly employ boundary detection operators such as the Sobel operator [52], Canny edge detection [53], and Laplacian of Gaussian (LoG) [54] to extract edge information from images. To further enhance segmentation accuracy, scholars have incorporated machine learning techniques, including random walks [55] and boundary neural fields [56], for refined boundary processing. Although the above methods enhance the extraction of boundary features, they may struggle with the high variability and complexity of spatial patterns in RS imagery, leading to inaccuracies in boundary delineation.

Recent advancements in applying deep learning methods to boundary enhancement of RS images semantic segmentation have yielded significant benefits, which fall into two main categories: refining loss functions and introducing auxiliary boundary branching. In the first strategy, optimizing the loss function encourages the segmentation model to prioritize edge precision during training. For instance, SBANet [57] introduces a boundary loss for improved boundary prediction and adaptively combines it with the original semantic loss, thereby improving segmentation outcomes and network generalization. To address the imbalance of RS images, FactSeg [58] employs cooperative probability, while MCCANet [59] leverages a joint boundary-class loss to enhance focus on foreground boundaries. The second strategy is to introduce auxiliary branching, where a separate subnetwork for boundary detection accurately determines the category of boundary pixels. For example,

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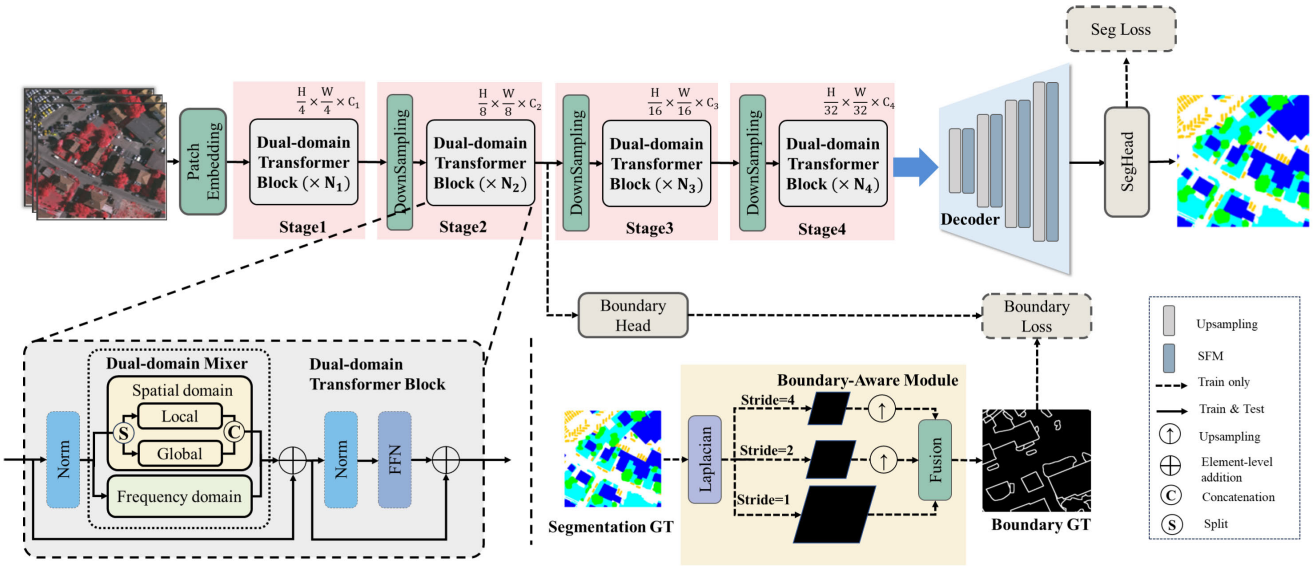


Fig. 2. Overall architecture of dual-domain transformer. The dual-domain transformer achieves high segmentation accuracy by extracting feature information in the frequency and spatial domains at the encoding stage. In addition, it utilizes a boundary-guided training strategy by utilizing the BAM to facilitate edge localization and segmentation capacity.

Jung et al. [60] adopt the holistically nested edge detection (HED) [61] approach to design additional auxiliary branches for extracting edge features at the encoder stage. Similarly, BESSL-Net [61] and BSNet [51] incorporate an auxiliary boundary extraction module in the encoding stage to extract edge margin information effectively.

Nevertheless, the above methods have limited segmentation performance and effectiveness on RS images, primarily owing to two major issues. First, the boundary extraction methods are intricate, leading to excessive parameters and computations. Second, these methods are limited to single-scale extraction at one layer, overlooking the diverse sizes of foreground categories typical in RS imagery. To address these concerns, we introduce a boundary-guided training strategy integrating a multiscale boundary extraction module. This module efficiently extracts multiscale boundary information as an auxiliary task, enhancing semantic segmentation with minimal parameter overhead.

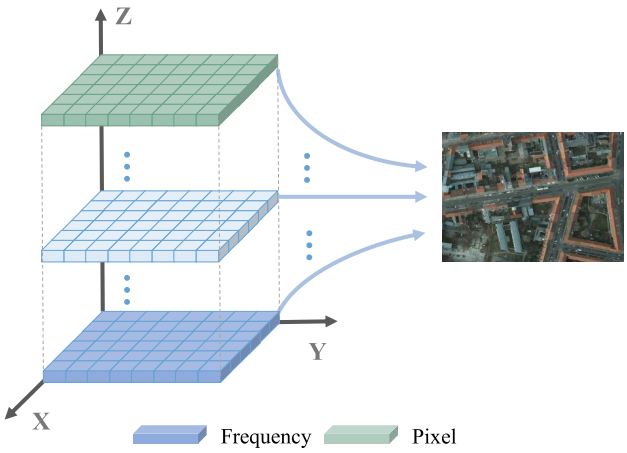


Fig. 3. Spatial–frequency domain relationship, where X and Y denote spatial dimensions, and Z signifies frequency domains. Each blue layer represents frequency components obtained by merging spatial pixels.

III. Method

In this section, we introduce dual-domain transformer, a spatial–frequency dual-domain transformer designed for RS image segmentation. The overall architecture is depicted in Fig. 2, followed by an in-depth discussion of the specifics, including the frequency-domain Fourier representation, overall architecture and essential components like dual-domain transformer block (DTB), the boundary-guided training strategy, and SFM.

A. Frequency-Domain Fourier Representation

The discrete Fourier transform (DFT), a vital element in our dual-domain transformer, is crucial for analyzing the frequency domain components of images and finds extensive use in various vision tasks [24], [47], [62]. Indeed, the DFT offers two primary advantages. One is that DFT simplifies spatial domain convolution operations into Fourier domain multiplications, allowing for a reduction in computational complexity. Another is its ability to dissect signals into constituent frequencies, allowing the extraction of image-size receptive fields from the frequency domain and revealing hidden features imperceptible in the spatial domain. As depicted in Fig. 3, the interplay between the spatial and frequency domains demonstrates that each frequency component is synthesized from aggregating all pixels in the spatial domain. This observation suggests that effectively modeling the frequency domain equates to capturing the essence of the spatial domain, highlighting the integral connection between these two aspects. Formally, the Fourier transform of a given image is denoted as

$$F(u,v) = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} f(x,y) \cdot e^{-2i\pi\left(\frac{x}{H}u + \frac{y}{W}v\right)} \tag{1}$$

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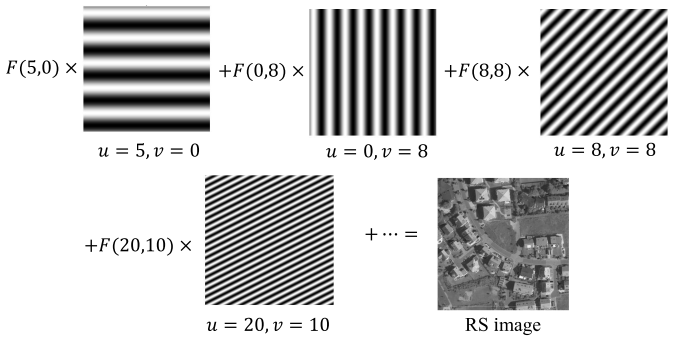


Fig. 4. Visualization of RS images in relation to 2-D sinusoidal frequency components.

where H and W denote the height and width of the image, the coordinates (x, y) and (u, v) correspond to spatial and frequency domains, with $f(x, y)$ and $F(u, v)$ representing their respective pixel and frequency domain values, respectively. Note that e represents the Euler number, and i denotes the imaginary unit, as defined by Euler's formula [63]

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{2}$$

thus, the natural exponential function in (1) can be rewritten as

$$e^{-i2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right)} = \cos 2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right) - i\sin 2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right). \tag{3}$$

According to (1) and (3), it can be seen that a Fourier transform dissects an image into orthogonal sine and cosine functions, generating the real and imaginary parts of the frequency domain. Conversely, these components represent the image's horizontal and vertical frequencies, each corresponding to a spatial domain sinusoidal wave. For a more intuitive grasp, we visualize the correlation between image frequency and its spatial domain sine wave representation. Illustrated in Fig. 4, the frequency domain coordinates (u, v) represent sinusoids with different angles and densities, with $F(u, v)$ signifying the contribution weight of each sinusoid to the image. Thus, the image can be composed of the product of numerous sinusoids with their contribution weights.

According to the DFT form, there are two fundamental elements: amplitude and phase. The amplitude specifies the strength or magnitude of a particular 2-D sinusoidal wave, and the phrase indicates the position of the sinusoidal wave relative to a reference point. Together, amplitude and phase allow for the complete reconstruction of the original image from its frequency components, with amplitude affecting the contrast and phase preserving the essential geometry.

As a variant of DFT, the FFT algorithm efficiently computes the DFT, significantly reducing the computational complexity and time, making it particularly advantageous for handling large datasets like RS imagery. Its counterpart, the inverse fast Fourier transform (IFFT), performs the reverse operation by converting frequency domain data back into the time or spatial domain. Collectively, FFT and IFFT enhance the efficiency of frequency component analysis.

B. Overall Architecture

Fig. 2 illustrates the architectural design of the dual-domain transformer, adhering to a conventional U-shaped encoder–decoder network design. Initially, the input image $X \in \mathbb{R}^{H\times W\times 3}$ is tokenized through a patch embedding layer which comprises 3×3 convolutions with a stride of 2, yielding a feature map of $(H/4) \times (W/4)$ resolution and C channels. The encoder backbone then sequentially progresses through four stages, each comprising a downsampling layer (a 3×3 stride-2 convolution) and multiple DTBs. However, the exception is the first stage, which includes only multiple DTBs. For consistency with conventional backbones, the feature map size at stage i is $(H/2^{i+1}) \times (W/2^{i+1})$, with $(C/2^{i-1})$ channels. To comprehensively capture image features while minimizing computational complexity, the encoder employs multiple DTBs (3, 3, 9, and 3 for $N_1 \sim N_4$ in stages 1–4, respectively). Each stage contains two parallel frequency-domain and spatial-domain branches to supplement feature extraction. As shown in Fig. 5, the spatial-domain branch is split into two paths along the channel, with the global-spatial branch using a self-attention mechanism to extract global features and the local-spatial branch using depthwise convolution and maxpool to extract local details, respectively. Meanwhile, the Fourier frequency branch exploits FFT to acquire an image-size receptive field for feature extraction. The final three branches concatenate together, and features are fused through a fusion module (consisting of a depthwise convolution and a linear layer). This dual-domain strategy ensures comprehensive feature capture by integrating spatial and frequency domains.

To enhance object boundary segmentation in RS imagery, we propose a boundary-guided training strategy utilizing BAM. This strategy leverages boundary details extracted from the encoder's second-stage feature maps (following [64]), which are rich in boundary information. The BAM module extracts multiscale boundary information from the segmentation ground truth (GT) to make the boundary GT. During training, a boundary head is inserted after the second stage. It utilizes boundary GT to optimize boundary loss, ensuring more accurate boundary delineation.

After the four stages of the encoder, the resultant feature map is transferred to the decoder. Like the encoder, the decoder comprises four stages to progressively restore the feature map to its original dimensions. Each stage comprises an upsampling module and an SFM. Following the Unet [15] architecture, we incorporate skip connections to facilitate detailed information transfer. Within the decoder, SFM adaptively integrates encoder and decoder features using an attention mechanism. Ultimately, the feature map is fed to SegHead to generate the final segmentation map. Throughout the training process, both seg loss and boundary loss are employed jointly to supervise the model.

C. Dual-Domain Transformer Block

Fig. 5(a) displays the main structure of DTB, aligning with the architecture of ViT but with a crucial modification: the replacement of ViT's multihead self-attention (MSA)

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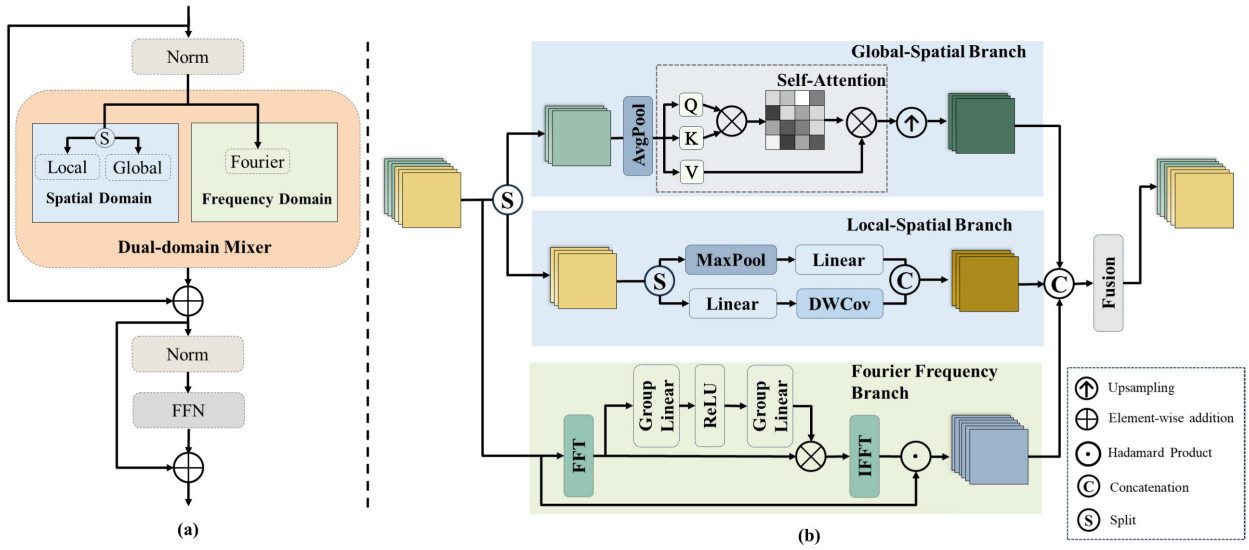


Fig. 5. (a) Overall architecture of the DTB and (b) specific details of the DualM. Each DualM in the DTB comprises spatial and frequency domain branches. The frequency domain branch is termed the Fourier Frequency branch, while the spatial domain branch is split into global and local branches, called the global-spatial and local-spatial branches, respectively.

mechanism with our proposed DualM for feature extraction. The FreT block can be represented as follows:

$$\hat{\boldsymbol{Y}} = \boldsymbol{X} + \text{DualM}(\text{LN}(\boldsymbol{X}))$$
$$\boldsymbol{Y} = \hat{\boldsymbol{Y}} + \text{FFN}(\text{LN}(\hat{\boldsymbol{Y}})) \tag{4}$$

where $\boldsymbol{X}$ and $\boldsymbol{Y}$ stand for the input and output of DTB, LN stands for layer normalization, FFN stands for feed-forward network, and DualM stands for DualM.

1) Dual-Domain Mixer: To comprehensively capture image features, we introduce a DualM consisting of spatial-domain and frequency-domain branches. As depicted in Fig. 5(b), the DualM partitions the input feature map into two branches. The spatial-domain branch splits along the channel dimensions, feeding into the global-spatial branch and local-spatial branch to extract global and local information efficiently. Concurrently, the Fourier frequency branch utilizes FFT to encompass the entire frequency spectrum, effectively perceiving image-size features from the entire spectrum.

Specifically, a given feature map $\boldsymbol{X} \in \mathbb{R}^{N\times C}$ is split along the channel dimension into $\boldsymbol{X}_g \in \mathbb{R}^{N\times C_g}$ and $\boldsymbol{X}_l \in \mathbb{R}^{N\times C_l}$, where $C_g + C_l = C$. Concurrently, X is denoted as $\boldsymbol{X}_f$ for input to the Fourier frequency branch. Subsequently, $\boldsymbol{X}_g$, $\boldsymbol{X}_l$, and $\boldsymbol{X}_f$ are fed into the global-spatial branch, local-spatial branch, and Fourier frequency branches, respectively, enabling comprehensive feature extraction. Noted, we employ the channel split mechanism. Within the four encoder stages of the DTB, C_l, and C_g are split at ratios of 2:1, 1:1, 3:7, and 1:11, progressively increasing the allocation to the global-spatial branch. These ratios align with the model's inclination to extract local detail information in shallow networks and primarily semantic information in deep networks.

2) Global-Spatial Branch: The global component, rich in semantic information, is crucial for object classification in RS images. Recent studies indicate that average pooling [50] and global attention [49] in MSA can effectively capture such global information. Therefore, we initially apply an average pooling layer, which reduces the resolution size of the feature maps to reduce the computational effort and enables the extraction of global information. Subsequently, MSA is employed to further extract global information in the feature map, which helps capture semantic information. Finally, an upsampling layer is employed to recover the feature map resolution after MSA. The global-spatial branch can be represented as follows:

$$\boldsymbol{Y}_g = \text{Upsample}\big(\text{MSA}\big(\text{AvgPool}\big(\boldsymbol{X}_g\big)\big)\big) \tag{5}$$

where $\boldsymbol{Y}_g$ represents the output of the global-spatial branch, MSA represents MSA, and AvgPool represents average pooling.

3) Local-Spatial Branch: Considering the detailed perception capabilities of MaxPool and the local feature extraction strength of CNN, we design a parallel bi-branch local-spatial branch. Specifically, We divide the input $\boldsymbol{X}_l$ into $\boldsymbol{X}_{l1} \in \mathbb{R}^{N\times(C_l/2)}$ and $\boldsymbol{X}_{l2} \in \mathbb{R}^{N\times(C_l/2)}$ along the channel dimension. $\boldsymbol{X}_{l1}$ undergoes a MaxPool and linear layer, while $\boldsymbol{X}_{l2}$ is processed through a linear layer and a depthwise convolution layer. Specifically, for the two subbranches, the channels are divided equally at a 1:1 ratio. Finally, we concatenate the outputs of the two branches. The process is expressed as follows:

$$\boldsymbol{Y}_{l1} = \text{FC}(\text{MaxPool}(\boldsymbol{X}_{l1}))$$
$$\boldsymbol{Y}_{l2} = \text{DwConv}(\text{FC}(\boldsymbol{X}_{l2}))$$
$$\boldsymbol{Y}_l = \text{Concat}(\boldsymbol{Y}_{l1}, \boldsymbol{Y}_{l2}) \tag{6}$$

where $\boldsymbol{Y}_{h1}$ and $\boldsymbol{Y}_{l2}$ represent the two subbranch outputs, $\boldsymbol{Y}_l$ represents the entire local-spatial branch output, FC represents fully connected layer, MaxPool represents max pooling, Concat represents concatenation, and DwConv represents depthwise convolution.

4) Fourier Frequency Branch: Drawing on the advantages of frequency analysis, we employ FFT to extract image-size features, resource-efficiently complementing the spatial

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domain approach. In this branch, $\boldsymbol{X}_f$ is transformed from the latent spatial to the frequency domain by FFT. This way, we get the frequency representation of the different frequency components corresponding to the spatial positions. We then employ a lightweight, learnable mask for adaptive and efficient extraction of the entire frequency spectrum information. Subsequently, we revert the frequency representation to the original latent space via IFFT. In addition, a residual connection is integrated to mitigate gradient explosion risks during frequency domain learning.

The process is expressed as follows:

$$\boldsymbol{Y}_f = \boldsymbol{X}_f + \mathcal{F}^{-1}[\mathcal{G}(\mathcal{F}(\boldsymbol{X}_f)) \odot \mathcal{F}(\boldsymbol{X}_f)] \tag{7}$$

where $\boldsymbol{X}_f$ and $\boldsymbol{Y}_f$ denote the input and output of the Fourier frequency branch, $\mathcal{F}(\cdot)$ denotes the FFT, and $\mathcal{F}^{-1}(\cdot)$ denotes the IFFT. In particular, $\mathcal{G}(\mathcal{F}(\boldsymbol{X}_f)$ is the mask tensor learned from $\boldsymbol{X}_f$, which is utilized to adaptively enhance important frequency components during the learning process. $\mathcal{G}(\cdot)$ is mainly implemented by two sets of 1×1 linear convolutional layers interspersed with a RELU activation function.

Ultimately, the outputs from the three branches are concatenated along the channel dimension, followed by a fusion module to enhance information integration. This module predominantly employs depthwise convolution for information exchange and a linear layer for channel dimension adjustment, maintaining consistency with the original input

$$\begin{aligned} \boldsymbol{Y}_c &= \text{Concat}\big(\boldsymbol{Y}_g, \boldsymbol{Y}_l, \boldsymbol{Y}_f\big) \\ \boldsymbol{Y}_{\text{FreM}} &= \text{FC}(\boldsymbol{Y}_c + \text{DwConv}(\boldsymbol{Y}_c)) \end{aligned} \tag{8}$$

where $\boldsymbol{Y}_c$ stands for the intermediate result of concatenating the output of the three branches and $\boldsymbol{Y}_{\text{DualM}}$ stands for the output of DualM.

Addressing the high-resolution characteristics of RS images, we apply three optimization strategies to our proposed DualM for efficiency and reduced hardware demands. First, the number of DTBs in the four stages is minimized and set to 3, 3, 9, and 3, creating a more compact network. Second, recognizing the proficiency of shallow stages in local-feature extraction and deep stages in global-feature processing, we employ a ramp structure in iFormer [65], gradually increasing channel numbers for the global-spatial branch from shallow to deep stages while reducing channels in the local-spatial branch, thus lowering computational complexity. Finally, considering the computational complexity and memory demands of Fourier transform in image segmentation, we initially exclude it from the high-resolution feature map in the first stage, focusing only on global-spatial and local-spatial branches, and introduce an entire branch in the final three stages.

D. Boundary-Guided Training Strategy

We develop a boundary-guided training strategy to enhance the model's segmentation precision, especially for edge details. This strategy integrates a BAM into the backbone network, guiding the low-dimensional layer to learn and capture edge details more effectively. As illustrated in Fig. 6(a), we initially apply multiscale Laplace kernels (2-d convolution kernels with different strides of 1, 2, and 4) to the segmentation GT for extracting multiscale boundary information. We then standardize resolution through up-sampling and achieve fusion via 1×1 convolution. Finally, a threshold of 0.1 is adopted to transform predicted details into the final binary boundary GT. Following this, a boundary head, depicted in Fig. 6(b), is inserted after stage 2 (following [64]) to generate boundary feature maps. We optimize the model training process to be boundary-aware by incorporating a boundary loss function. Specifically, the boundary GT is used as an optimization target for the feature maps generated by the boundary head. This approach simplifies the model's boundary guidance, resulting in more accurate segmentation of boundary details without introducing additional computational costs in the inference process.

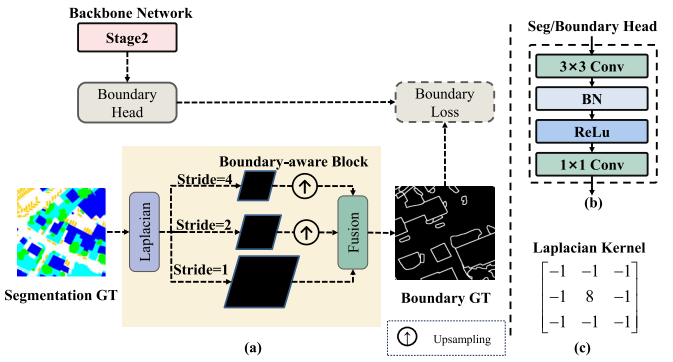


Fig. 6. Detailed depiction of the boundary-guided training strategy employing BAM. (a) Overall architecture of the BAM. (b) Boundary head in the BAM, consistent with the seg head structure. (c) Laplacian kernel in the BAM.

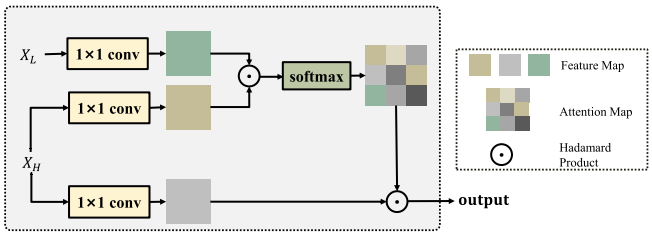


Fig. 7. Detailed view of SFM. SFM mainly uses 1×1 convolution and the attention mechanism to adaptively and efficiently fuse the features $\boldsymbol{X}_L$ from the encoder and $\boldsymbol{X}_H$ from the decoder.

E. Scale-Feature Fusion Module

In our decoder, we designed an SFM to integrate the encoder's low-level details efficiently with the decoder's features. Utilizing a compact 1×1 self-attention mechanism, the SFM dynamically assigns different weights, selectively enhancing data from the encoder or decoder. As illustrated in Fig. 7, the SFM is integral to each stage of the decoder, accepting two inputs: the low-level feature map $\boldsymbol{X}_L$ from the encoder and the high-level counterpart $\boldsymbol{X}_H$ from the decoder. The high-level input, $\boldsymbol{X}_H$, is split into two branches, while $\boldsymbol{X}_L$ follows a single branch, each undergoing a 1×1 convolution. The outputs of these branches interact, with one $\boldsymbol{X}_H$ branch output undergoing a Hadamard product with the $\boldsymbol{X}_L$ output, creating an attention map via Softmax activation. This attention map then merges with the feature map from the other $\boldsymbol{X}_H$ branch to

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produce the final output. The SFM facilitates efficient, adaptive fusion of low- or high-dimensional information, enhancing the overall decoding process. This process can be summarized as follows:

$$\boldsymbol{Y}_o = \sigma[\text{Conv}(\boldsymbol{X}_L) \odot \text{Conv}(\boldsymbol{X}_H)] \odot \boldsymbol{X}_H \tag{9}$$

where $\boldsymbol{Y}_o$ represents the output of the SFM, σ stands for softmax activation function, Conv stands for 1×1 convolution, and $\odot$ stands for Hadamard product.

IV. Experiments

A. Datasets

1) Vaihingen: The Vaihingen dataset [66] contains 33 urban images with very-high-resolution of 2496×2064 pixels, covering near-infrared, red, and green bands. This dataset presents unique challenges due to its urban landscape, characterized by small, scattered villages and objects that shade each other. The dataset includes one background class, Clutter/background, and five foreground classes: impervious surface, tree, low vegetation, building, and car. In our experiments, following [11], [21], we designate 16 images (with selected image IDs of 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, 37) as the training set, with the remaining 17 allocated to the test set.

2) Potsdam: The Potsdam dataset [67] includes 38 RS images, each with dimensions of 6000×6000 pixels. According to the city characteristics, the images in the Potsdam dataset have extensive building blocks, narrow streets, and densely gathered structures of buildings, with the same foreground background categories as Vaihingen. Notably, the dataset comprises images in red, green, blue, and NIR spectral bands, along with DSM and NDSM data. In our experiments, we focus on the RGB bands. Following [11], [21], we use 24 images for training (identified by selected image IDs: 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_11, 7_12, 7_13) and the remaining 14 for testing.

3) Zeebrugge: The Zeebrugge dataset [68] comprises seven very-high-resolution RS images, each with dimensions of $10\,000 \times 10\,000$ pixels. Situated in the city and its surrounding harbor, Zeebrugge provides images rich in semantic detail, categorized into eight classes: boats, impervious surface/road, low vegetation, trees, buildings, cars, clutter, and water. This dataset, derived from the 2015 IEEE GRSS Data Fusion Contest, provides semantic labels for five training images, while two unlabeled images (tiles 4 and 6) serve as the test set.

B. Implementation Details

1) Training Settings: Our experiments use Python 3.9 and Pytorch 1.13.0 on a Windows 10 operating system, with model training executed on a single NVIDIA GTX 3090ti GPU. During training, we utilize AdamW optimizer (as recommended in [69]) with an initial learning rate of $6e^{-}4$, following the configuration suggested in [11]. The learning rate is modulated using a cosine strategy to expedite convergence. To augment data diversity and thereby enhance training efficacy, a random cropping approach is adopted, resizing images to 512×512 pixels. In addition, we incorporate several data augmentation techniques, such as random scaling, vertical flipping, and horizontal flipping, across a training duration of 200 epochs for enhanced model performance.

2) Loss Functions: Fig. 8 illustrates the proportion of each semantic category in the three datasets, Vaihingen, Potsdam, and Zeebrugge. Analysis reveals an imbalance in the proportion of each semantic category, with some foreground categories having a small proportion while others dominate. This imbalance can lead to a learning bias during training, favoring the optimization of categories with a larger share, such as "impervious surface," while neglecting smaller ones, such as "car" in the Vaihingen and Potsdam datasets. To mitigate this issue, we implement two essential corrections. First, following the methodologies in [11] and [21], we exclude the "clutter/background" category, which disproportionately influences loss calculation and model evaluation. Second, we incorporate the dice loss [70] L_{dice} into the loss function, which is highly effective for tasks with unevenly distributed categories. Combining it with the cross-entropy loss L_{ce}, we form the total joint loss L, optimizing the model more effectively. In addition, edge pixels in an object are significantly fewer than nonedge pixels, making edge prediction a problem of unbalanced category distribution. To address this, we employ binary cross-entropy loss, which is suitable for the binary nature of edge categorization, and dice loss for joint optimization in edge learning. The total loss L_{total} can be formulated explicitly as follows:

$$\begin{aligned} L_{\text{total}} &= L_{\text{seg}} + \alpha \times L_{\text{bound}} \\ L_{\text{seg}} &= L_{\text{dice}_s} + L_{\text{ce}_s} \\ L_{\text{bound}} &= L_{\text{dice}_b} + L_{\text{bce}_b} \end{aligned} \tag{10}$$

where L_{seg} and L_{bound} denote segmentation and boundary detection losses, respectively. L_{bound} is multiplied by a hyperparameter, α, which represents its proportion in the total loss. In our experiments, α is set to 0.5. Specifically, the segmentation loss L_{seg} is composed of the dice loss L_{dice_s} and cross-entropy loss L_{ce_s}. Similarly, the boundary loss L_{bound} comprises the dice loss L_{dice_b} and binary cross-entropy loss L_{bce_b}.

3) Evaluation Metrics: In our experiments, we employ two primary types of evaluation metrics to assess the model's performance. The first focuses on segmentation accuracy, utilizing metrics like average $F1$ score (avg.F1), overall accuracy (OA), average accuracy (AA), and mean intersection over union (mIoU). The second evaluates model efficiency and size, considering floating point operations per second (FLOPS) for model complexity and the number of model parameters (measured in M) for memory requirements.

C. Comparison With SOTA Approaches

To evaluate the efficacy of our dual-domain transformer, we conduct comparative analyses with predominant semantic segmentation networks, which include CNN-based methods (e.g., Unet [17], ShelfNet [71], FANet [72], EaNet [73], ABCNet [74], BANet [75], MANet [4], CCANet [76],

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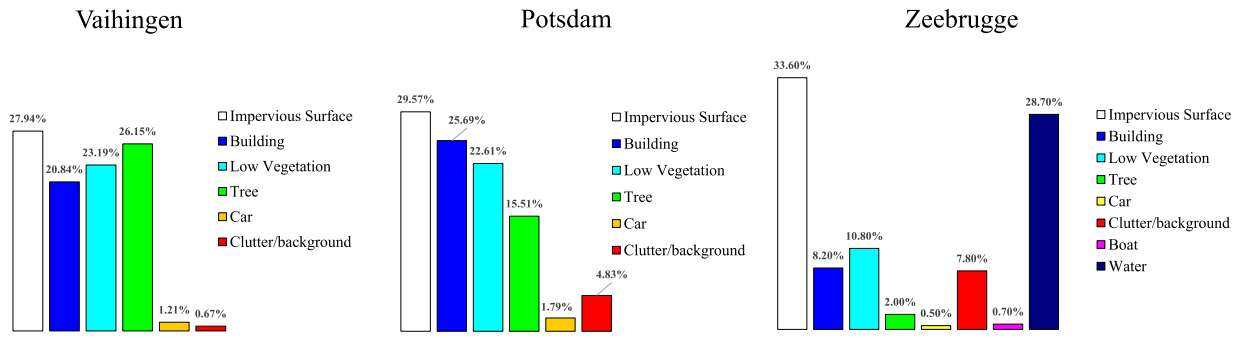


Fig. 8. Imbalanced semantic category distribution in the Vaihingen, Potsdam, and Zeebrugge datasets.

TABLE I
COMPARATIVE ANALYSIS OF SEGMENTATION RESULTS OBTAINED BY DUAL-DOMAIN TRANSFORMER AND EXISTING METHODS USING THE VAIHINGEN DATASET

Methods	F1 (%)					Evaluation Metrics		
	Impervious Surface	Building	Low Vegetation	Tree	Car	Avg.F1 (%)	OA (%)	mIoU (%)
Unet (2015) [17]	89.5	91.7	80.9	88.2	75.0	85.1	87.6	74.5
ShelfNet (2019) [71]	91.8	94.6	83.8	89.3	77.9	87.5	89.8	78.3
FANet (2020) [72]	90.7	93.8	82.6	88.6	71.6	85.4	88.9	75.6
EaNet (2020) [73]	91.7	94.5	83.1	89.2	80.0	87.7	89.7	78.7
ABCNet (2021) [74]	90.5	92.7	81.4	88.2	76.3	85.8	88.3	75.6
BANet (2021) [75]	90.8	93.7	82.4	88.9	79.3	86.9	88.9	77.2
MANet (2021) [4]	92.2	95.1	83.2	89.3	87.9	89.5	90.2	81.3
Segmenter (2021) [18]	89.8	93.0	81.2	88.9	67.6	84.1	88.1	73.6
BoTNet (2021) [78]	89.9	92.1	81.8	88.7	71.3	84.8	88.0	74.3
STUNet (2022) [21]	91.4	94.1	83.2	89.1	82.6	88.1	88.6	77.2
DCSwin (2022) [10]	90.3	92.0	81.6	88.1	77.7	86.0	88.1	75.8
UnetFormer (2022) [11]	92.4	**95.6**	83.5	89.5	87.4	89.7	90.5	81.6
EMRT (2023) [79]	89.2	92.7	78.7	86.5	72.1	83.8	-	75.0
CGVT-S (2023) [40]	92.0	95.2	82.6	89.2	87.5	89.3	91.2	-
Ours (Dual-domain Transformer)	**93.2**	95.5	**84.6**	**89.8**	**88.9**	**90.4**	**91.4**	**82.7**

and MFNet [77]), transformer-based methods (e.g., Segmenter [18], DCSwin [10], and CGVT [40]), and hybrid CNN-transformer models (e.g., BoTNet [78], STUNet [21], UnetFormer [11], and EMRT [79]). It is worth noting that Unet, ShelfNet, FANet, and Segmenter are benchmarks in general semantic segmentation, while the others are specifically tailored for RS imagery segmentation. For EMRT, CGVT, MFNet, and CCANet, we utilize data from the original paper (results not provided in the original papers are denoted as "-"). For other networks, we rely on results reproduced within this experimental setting. Specifically, for CGVT, we compare its lightweight version, CGVT-S, considering the same network scale as our dual-domain transformer. In all tables, **bolded** numbers signify optimal results, while underlined ones denote suboptimal outcomes.

1) Assessing Semantic Segmentation Performance Using the Vaihingen Dataset: Table I illustrates the semantic segmentation results for the mentioned methods on the Vaihingen dataset [66]. Overall, our dual-domain transformer achieves the best results, with avg.F1 reaching 90.4%, OA reaching 91.4%, and mIoU reaching 82.7%. Notably, dual-domain transformer achieves optimal segmentation accuracy on most categories, especially those with small interclass differences and challenging boundaries, such as "tree" and "low vegetation." These results underscore the superior semantic segmentation capabilities of our dual-domain transformer for RS imagery.

Fig. 9 displays part of the visualization results, observing that CNN-based methods yield less satisfactory segmentation than transformer-based methods, often resulting in incomplete object semantics and fragmented segments. For instance, methods like Unet and ABCNet struggle to accurately segment structures like "building" in the third and fourth rows. In contrast, transformer-based methods, particularly those combining CNN and transformer, show notable improvements in addressing these challenges. However, the hybrid methods still encounter difficulties in precise boundary localization and correctly classifying pixels. Our proposed dual-domain transformer surpasses these challenges, achieving superior accuracy in categorization and enhancing boundary segmentation precision compared to other methods.

2) Assessing Semantic Segmentation Performance Using the Potsdam Dataset: Comparative experiments are conducted on the Potsdam dataset [67] to substantiate the efficacy and generalizability of our proposed dual-domain transformer. As Table II demonstrates, dual-domain transformer surpasses other methods, achieving the highest avg.F1 of 92.4% and outperforming the second place by 1.1%. Generally, the segmentation accuracy on the Potsdam dataset surpasses that of Vaihingen because the objects in the Potsdam are dispersed, and the segmentation difficulty is relatively low. Despite transformer-based methods' global modeling capabilities, they do not outperform CNN-based attention methods. This is credited to the objects occurring with each other with fewer

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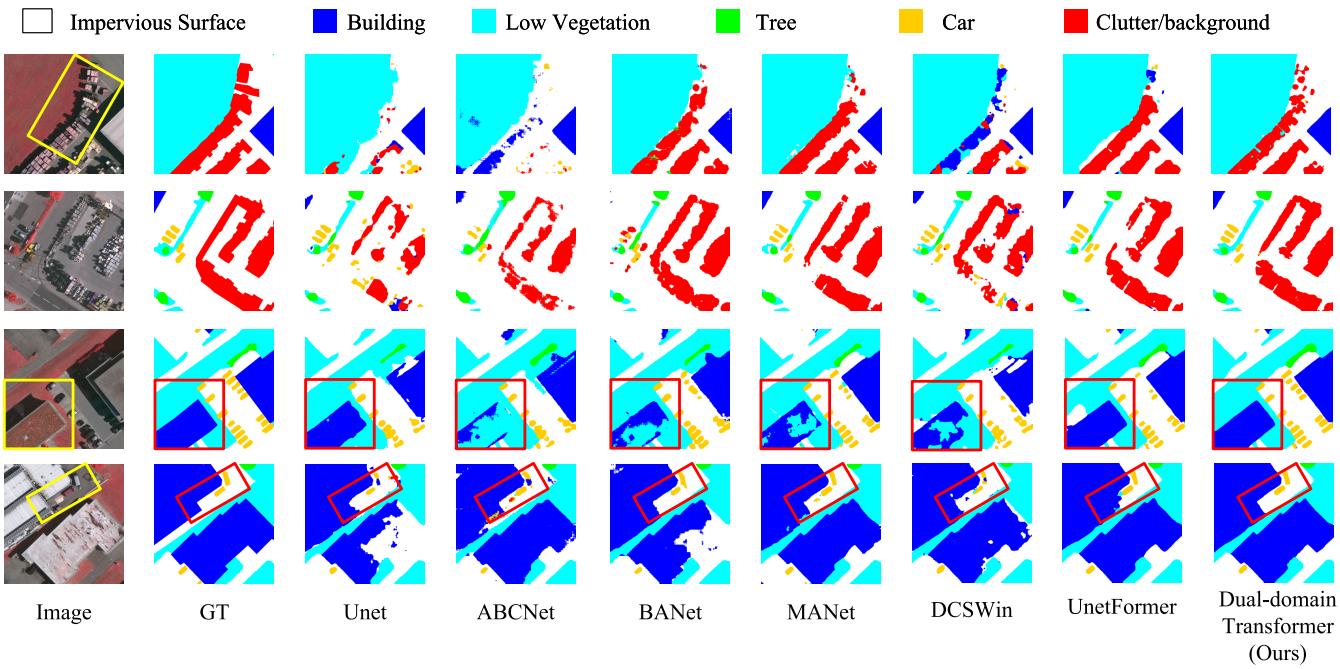


Fig. 9. Comparative visualization of semantic segmentation predictions on the Vaihingen dataset.

TABLE II
COMPARATIVE ANALYSIS OF SEGMENTATION RESULTS OBTAINED BY DUAL-DOMAIN TRANSFORMER AND EXISTING METHODS USING THE POTSDAM DATASET

Methods	F1 (%)					Evaluation Metrics		
	Impervious Surface	Building	Low Vegetation	Tree	Car	Avg.F1 (%)	OA (%)	mIoU (%)
Unet (2015) [17]	87.8	90.1	82.6	81.7	92.5	87.1	84.9	77.4
ShelfNet (2019) [71]	92.5	95.8	86.6	87.1	94.6	91.3	89.9	84.4
FANet (2020) [72]	92.0	96.1	86.0	87.8	94.5	91.3	89.8	84.2
EaNet (2020) [73]	92.0	95.7	84.3	85.7	95.7	90.6	88.7	83.4
ABCNet (2021) [74]	87.7	91.7	81.6	79.9	90.4	86.2	84.5	76.1
BANet (2021) [75]	90.8	94.2	84.7	84.8	93.3	89.6	87.9	81.3
Segmenter (2021) [18]	91.5	95.3	85.4	85.0	88.5	89.2	88.7	80.7
BoTNet (2021) [78]	92.6	95.3	86.5	87.5	89.7	90.3	88.9	83.8
DCSwin (2022) [10]	89.3	92.9	83.6	83.8	92.9	88.5	86.7	79.6
UnetFormer (2022) [11]	91.8	95.4	86.3	87.5	95.4	91.3	89.7	84.2
EMRT (2023) [79]	90.9	94.9	84.1	85.2	90.3	89.1	-	80.5
Ours (Dual-domain Transformer)	**93.2**	**97.0**	**86.9**	**88.8**	**95.9**	**92.4**	**90.8**	**86.0**

semantic ambiguities, and good segmentation performance can be achieved using only CNN-based methods. Remarkably, our dual-domain transformer outperforms $F1$ accuracy across all five foreground categories, primarily excelling in the "tree" category. This enhanced performance is attributed to the dual-domain transformer's capability to decouple feature information in the frequency domain, enabling it to distinguish effectively between visually similar objects such as "tree" and "low vegetation."

The visual results presented in Fig. 10 demonstrate the superiority of dual-domain transformer in segmentation over other methods, supported by three crucial observations. First, the dual-domain transformer achieves accurate segmentation, notably in localizing "building" in the lower part of the first row, surpassing the other methods. Second, the dual-domain transformer exhibits improved precision and boundary smoothness, notably in the challenging segmentation of "building" and "low vegetation" in the third row. Moreover, the dual-domain transformer effectively differentiates the "building" boundaries in the fourth row, unlike other methods that blur them.

3) Assessing Semantic Segmentation Performance Using the Zeebrugge Dataset: The Zeebrugge dataset is sourced from the GRSS Data Fusion Contest [68], with the official GT of the test set remaining unpublished. Consequently, we need to submit our experimental outcomes to the official website to access the results online. The contest uses AA and OA as evaluation metrics. Accordingly, in Table III, we evaluate the model's segmentation performance based on these metrics. Overall, our dual-domain transformer achieves the best segmentation results, with AA at 82.06% and OA at 88.95%. Specifically, our dual-domain transformer achieves the best in five categories, including "impervious surface," "clutter," "building," "tree," and "car."

Furthermore, we analyze the network performance by visualizing the results. Since the 2015 IEEE GRSS Data Fusion Contest does not provide GT, we can only compare and analyze the segmentation results with the original images.

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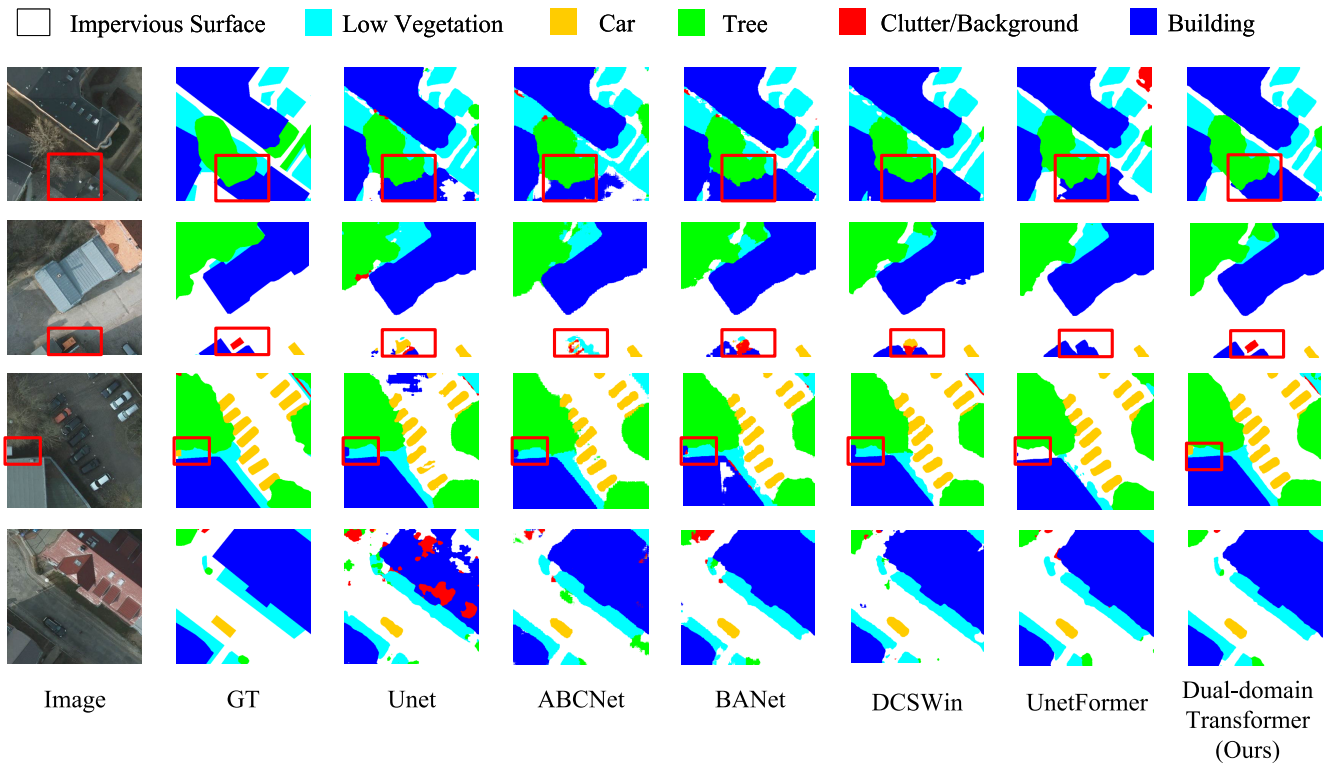


Fig. 10. Visual comparison of the semantic segmentation prediction results obtained from the Potsdam dataset.

TABLE III
COMPARATIVE ANALYSIS OF SEGMENTATION RESULTS OBTAINED BY DUAL-DOMAIN TRANSFORMER AND EXISTING METHODS USING THE ZEEBRUGGE DATASET

Methods	AA (%)								Evaluation Metrics	
	Impervious Surface	Water	Clutter	Low Vegetation	Building	Tree	Boat	Car	AA (%)	OA (%)
Unet(2015)	77.97	98.83	44.98	76.89	81.5	49.61	36.86	62.29	66.12	81.87
ABCNet(2021)	79.72	98.97	44.38	77.83	75.43	40.12	32.73	45.41	61.83	81.2
MANet(2021)	87.38	98.97	65.21	80.13	85.88	86.09	60.24	89.91	81.73	88.42
MFNet(2021)	86.95	**99.03**	61.72	81.77	84.55	77.52	69.29	88.65	-	88.87
CCANet(2021)	87.11	98.75	61.45	**85.68**	82.95	80.36	**70.25**	88.05	81.83	88.83
UnetFormer(2022)	84.31	98.98	66.55	83.66	64.02	88.72	60.04	88.23	79.31	85.37
Dual-domain Transformer	**89.37**	98.86	**66.87**	76.24	**86.74**	**88.96**	61.58	**89.93**	**82.06**	**88.95**

Initially, we observe the segmentation results of the whole image. As shown in Fig. 11, our dual-domain transformer exhibits smoother edges and fewer segmentation fragments than other networks. Subsequently, Fig. 12 shows the specific segmentation details of two test images (tiles 4 and 6). In the first row of tile 4, Unet, ABCNet, BANet, and UnetFormer inaccurately segment the docked "boats" accurately, and there are burrs on the edges of the ports. On the contrary, our dual-domain transformer can segment smooth edges relatively well and be more accurate in segmenting "boats." In the second row, the road at the crossing, categorized as "impervious surface/road," is very similar to the "building" and "low vegetation," making misclassifying it easy. A similar situation occurs in tile 6. In the top row, a soccer field covered with grass (categorized as "low vegetation") appears exposed due to foot traffic, leading ABCNet and BANet to misclassify it as "tree," and UnetFormer as "impervious surface/road." In the second row, the rounded portion in the middle of the road is misclassified as either "low vegetation" or "building." Notably, our dual-domain transformer can handle the above problems very well and reduce misclassification.

4) Analysis of Efficiency: When semantic segmentation of RS images is performed, the balance between model complexity and practicality is crucial. Models need to be sophisticated enough to extract comprehensive context information effectively, but not so complex that they become impractical for resource-limited environments like mobile or embedded devices. Generally, high-performance models often expand their receptive fields by incorporating attention mechanisms, such as transformer-based methods, thereby increasing their complexity. Conversely, CNN-based methods are less complex but often less effective. In evaluating the parameters and computational resource demands of the dual-domain transformer, our analysis in Table IV compares various representative models on memory storage needs (Params), computational complexity (FLOPS), and segmentation accuracy (OA). The results indicate that our dual-domain transformer has the best segmentation results (OA as high as 91.2%) with slightly

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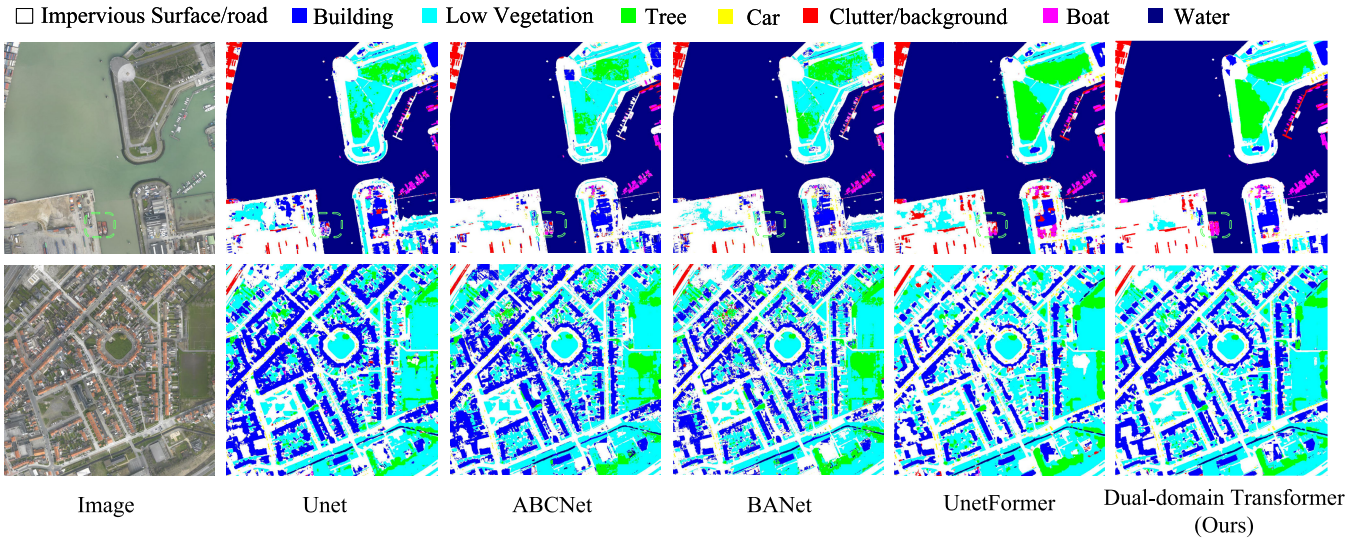


Fig. 11. Visual comparison of the semantic segmentation prediction results obtained from the Zeebrugge dataset.

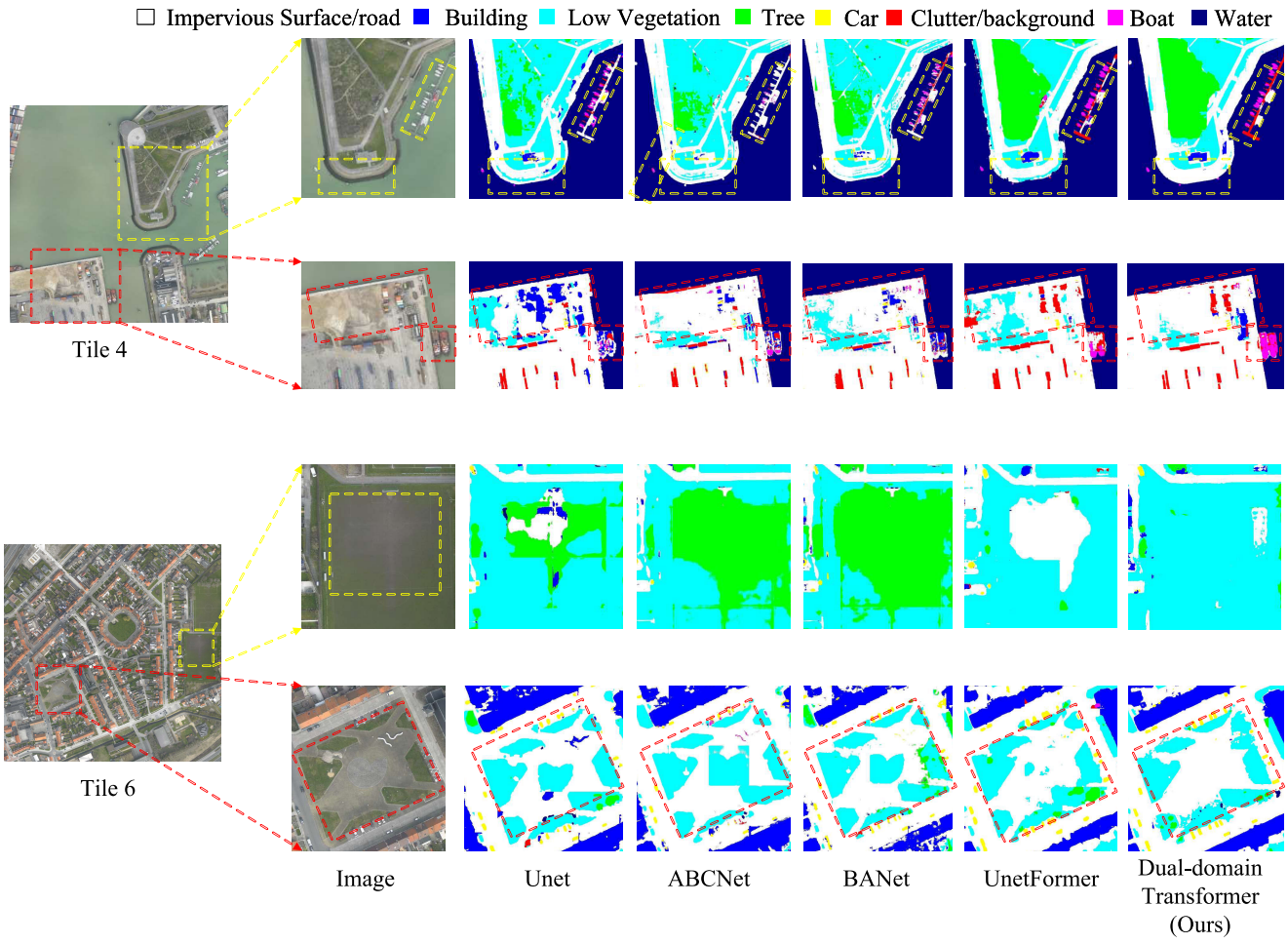


Fig. 12. Detailed visual comparison of the semantic segmentation prediction results obtained from the Zeebrugge dataset Tile 4 and 6.

more parameters than CNN-based methods such as ABCNet and BANet but considerably fewer than transformer-based models, indicating its less resource demands. Furthermore, our dual-domain transformer exhibits lower FLOPS, especially against recent Swin transformer-based models like STUNet and DCSwin, highlighting its efficiency. It is noteworthy that

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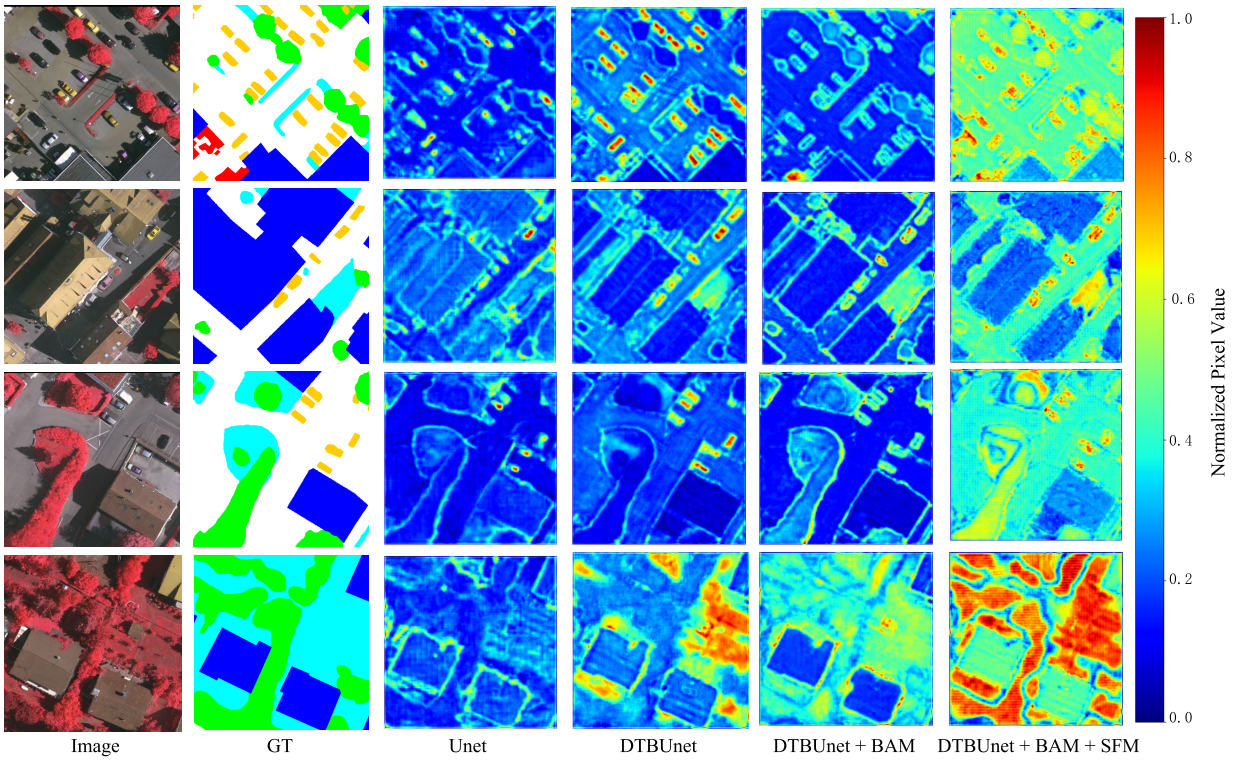


Fig. 13. Visualization of individual module feature maps (feature maps taken from the second last layer of the network). Compared to the baseline network Unet, DTBUnet pays more attention to information such as "car," with the BAM significantly boosting edge attention and the SFM augmenting focus for "impervious surface," "tree," and "low vegetation" categories.

TABLE IV
COMPARISON OF MODEL PARAMETERS, COMPLEXITY, AND SEGMENTATION ACCURACY ON THE VAIHINGEN DATASET

Methods	Params (M)	FLOPS (G)	OA (%)
Unet (2015) [17]	14.2	33.2	87.6
ABCNet (2021) [74]	14.0	31.3	88.3
BANet (2021) [75]	12.7	26.1	88.9
MANet (2021) [4]	35.9	155.5	90.2
STUNet (2022) [21]	161.0	214.8	88.6
DCSwin (2022) [10]	66.9	140.3	88.1
CGVT-S (2023) [40]	57.4	56.8	91.2
Dual-domain Transformer	33.9	76.1	**91.4**

CGVT comprises three variants: CGVT-S, CGVT-M, and CGVT-L. For comparison, we utilize CGVT-S, equivalent in volume to the dual-domain transformer. Our dual-domain transformer outperforms CGVT-S by 0.2%, with 23.5M fewer parameters and 19.3G higher FLOPS. In short, dual-domain transformer efficiently balances the tradeoffs between efficiency, memory usage, and computational complexity, while maintaining outstanding segmentation performance that comprehensively surpasses both CNN and transformer-based models.

D. Ablation Study

To validate the efficacy of the proposed DTB, SFM, and boundary-guided training strategy, we performed ablation experiments on the Vaihingen dataset.

1) Overall Analysis of Each Component: Building upon a baseline UNet model, we conduct ablation studies by sequentially introducing three components: 1) DTB; 2) SFM; and 3) boundary-guided training strategy, referred to as BAM instead. Table V presents the incremental enhancement in segmentation performance with each addition. Furthermore, the model achieves optimal performance with the three components used jointly, with a mIoU of 82.7%.

2) Ablation Experiment on the DTB: As seen in Table V, adopting DTB yielded significant improvement, boosting the mIoU by 7.6%, which validates its significance to the model. Among the five foreground categories, DTB achieves the most significant segmentation performance improvement for the "car" category, with an 18.5% increase in mIoU. This result is further supported by the feature map visualization in Fig. 13, where DTBUnet (incorporating the DTB in its encoder and employing the Unet architecture in its decoder) focuses more on the "car" category than Unet in the first row. Precisely because the strength of DTB lies in extracting complementary features across various levels and domains, resulting in a significant mIoU enhancement in the "car" category. In addition, Fig. 14 presents the visualization of the segmentation results. DTBUnet demonstrates more precise segmentation, effectively addressing the segmentation discontinuity issues, as observed in the first and second rows of the "building" category. Similarly, for the "car" category (third row), DTBUnet successfully handles shadow interference, distinctly segments each vehicle, and accurately delineates its boundaries.

We further assess each branch's impact on DualM. The following ablation experiments only change the branching structure inside the DualM, and the overall architecture

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TABLE V
EVALUATING DTB, BAM, AND SFM ON VAIHINGEN DATASET VIA ABLATION STUDIES WITHIN THE VAIHINGEN DATASET

Methods	IoU (%)					Evaluation Metrics		
	Impervious Surface	Building	Low Vegetation	Tree	Car	mIoU (%)	Param (M)	FLOPs (G)
Baseline Unet	81.1	84.8	68.1	78.9	60.1	74.6	14.2	33.2
+ DTB	86.7	91.7	72.6	81.4	78.6	82.2	35.2	80.7
+ DTB + BAM	86.7	91.8	72.5	81.4	79.2	82.3	35.2	80.7
+ DTB + SFM	86.7	91.8	72.2	80.8	**80.3**	82.4	33.8	76.1
+ DTB + SFM + BAM (Ours)	**87.2**	91.4	**73.3**	**81.5**	80.0	**82.7**	33.9	76.1

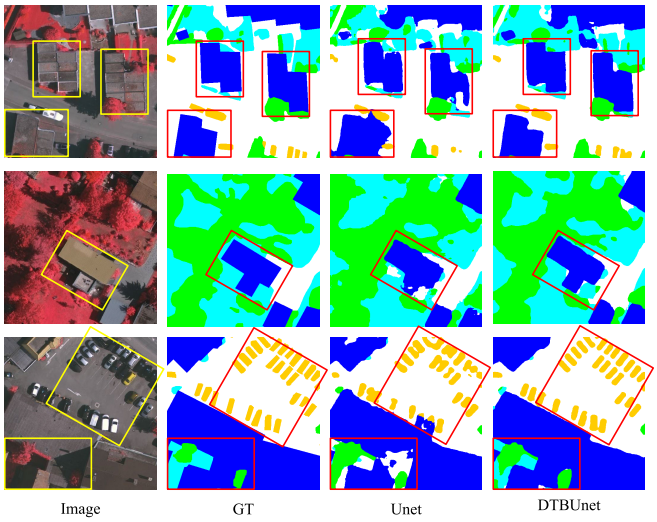


Fig. 14. Visualizing and comparing dual-domain transformer ablation experiment segmentation results.

remains unchanged. According to the channel split mechanism (mentioned in Section III-C), when using only one spatial-domain branch of DualM, all channel features are assigned to it. When both spatial-domain branches are utilized, features are divided along the channel dimensions at ratios of 2:1, 1:1, 3:7, and 1:11 for feeding into the global-spatial branches and the local-spatial branches in the four encoding stages, respectively. Similarly, in the local-spatial branch, the channels are split equally at a 1:1 ratio if both branches are used; otherwise, all features are assigned to the single branch. As illustrated in Table VI, using one single domain branch yields suboptimal performance. The most effective performance, with a mIoU of 79.4%, is achieved by employing solely the Fourier frequency branch, affirming its efficacy. Next, we combine the global-spatial branch with the local-spatial branch and Fourier frequency branch. These combinations enhance performance, achieving mIoU scores of 81.6% and 81.9%, respectively. Employing all three branches (the global-spatial, local-spatial, and Fourier frequency), the segmentation performance reaches the best (mIoU of 82.2%). Subsequently, by visualizing the feature maps from the fourth stage of the encoder (see Fig. 15), we observe that the global-spatial branch captures more information than the local-spatial branch. In addition, the Fourier frequency branch also provides comprehensive feature extraction and complements the spatial-domain branches. Our dual-domain transformer, which integrates all three branches, achieves a more comprehensive feature extraction and enhances image understanding.

3) Comparison of Networks Using Different Backbones: To validate the effectiveness of our proposed encoder DTB, we perform ablation experiments by replacing similar-size backbones. As shown in Table VII, we compare ResNet50 [80], Vit-S [34], and Swin-S [23] at the same scale. Results in Table VII demonstrate that our dual-domain transformer achieves superior segmentation accuracy, with an OA of 91.4%, which is 2.9% higher than ResNet50, 6.3% higher than Vit-S, and 3.9% higher than Swin-S. Regarding the number of parameters and computational complexity, dual-domain transformer is higher than ResNet50 but much lower than Vit-S and Swin-S. Overall, the dual-domain transformer is highly efficient, offering excellent segmentation performance with modest resource requirements.

4) Ablation Experiment on the SFM: Table V illustrates a slight performance improvement with the SFM integration, along with a reduction in parameter count by 1.4M and 4.6G FLOPS. Fig. 16 presents the visualization results, demonstrating enhanced segmentation performance by integrating SFM into DTBUnet. Specifically, in challenging scenarios such as in the first and second rows, where shadows from the tall "building" obscure lower "car" or other buildings, adding SFM markedly improves segmentation. This enhancement is due to SFM's capacity to convey detailed positional features from the encoder to the decoder, thereby facilitating accurate position recovery. Moreover, in the third row, SFM demonstrates effective localization and categorization of "clutter/background," showcasing its robustness in complex segmentation contexts.

5) Effects of the Boundary-Guided Training Strategy: We further explore the significance of the boundary-guided training strategy. Results in Table V demonstrate its effectiveness, yielding a 0.3% mIoU enhancement without a notable increase in parameters and FLOPS. Subsequently, in Fig. 17, we present visualizations including the original image, the GT, the DTBUnet + SFM Boundary (boundary map for the segmentation result using the DTBUnet model with a dual-domain transformer in the encoder and SFM in the decoder), and the DTBUnet + SFM + BAM Boundary (boundary map of the segmentation result after using the boundary-guided training strategy). Observing Fig. 17, we notice that the extracted boundaries become smoother and more consistent with GT after using the boundary-guided training strategy, particularly where different object categories intersect. In the first row, the boundary classification between the "car" and the "building" obscured by shadows is more accurate after using the boundary-guided training strategy. Similarly, in the second row, the segmentation of "tree" amidst "low vegetation" shows marked improvement with the boundary-guided training strategy.

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TABLE VI
ABLATION EXPERIMENTS ON DIFFERENT BRANCHES IN DUALM

Global-Spatial	Local-Spatial (Pool)	Local-Spatial (DW)	Fourier Frequency	Param (M)	FLOPS (G)	mIoU (%)
✓				25.3	61.3	78.0
	✓	✓		26.5	66.7	70.8
			✓	30.3	72.2	79.4
✓	✓			25.5	62.7	79.2
✓	✓	✓		25.4	62.1	81.6
✓			✓	32.2	75.4	81.9
✓	✓	✓	✓	35.2	80.7	**82.2**

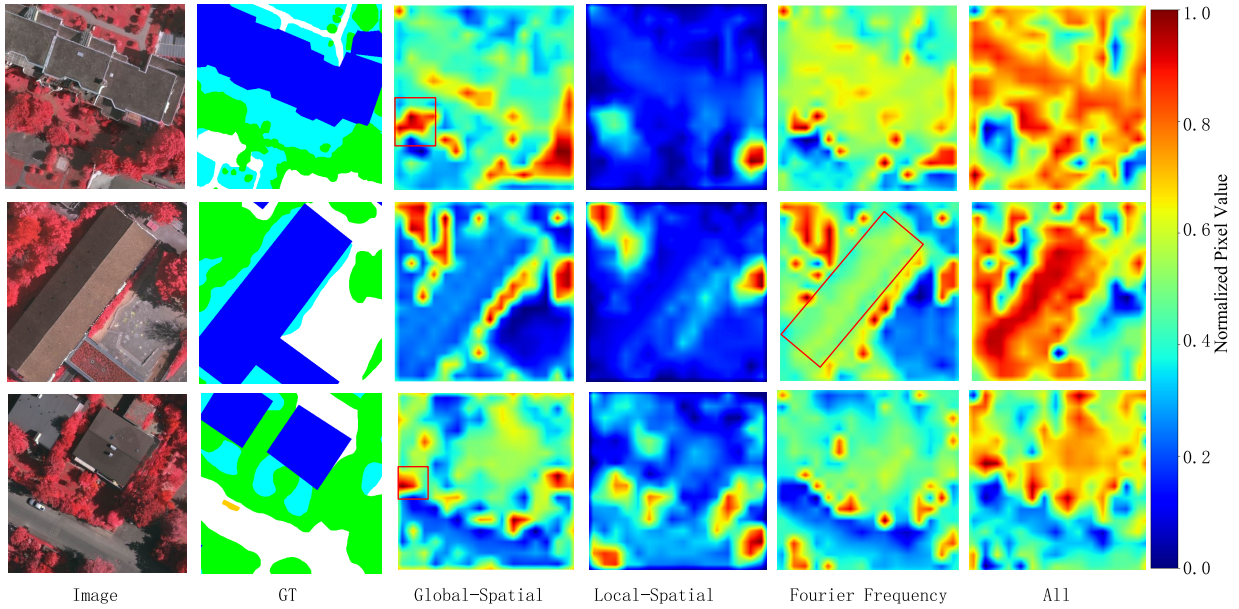


Fig. 15. Visualization of feature maps for different branches of DualM (feature maps taken from the last layer of encoder stage 4). The Fourier frequency branch and the spatial-domain branch can complement each other to extract features. For instance, in the first and third labeled rows, the global-spatial branch can perceive the corresponding features while the Fourier branch does not. In the second row, the Fourier can perceive the features the spatial-domain branch does not perceive. Our method uses three branches to achieve comprehensive coverage of the image.

TABLE VII
ABLATION EXPERIMENTS ON BACKBONE NETWORKS

Backbone	Params (M)	FLOPS (G)	OA (%)
ResNet50 [80]	24.7	46.6	88.5
Vit-S [34]	45.5	85.4	85.1
Swin-S [23]	62.2	101.4	87.5
Dual-domain Transformer	33.9	76.1	**91.4**

6) Visualization of Individual Module Feature maps: Fig. 13 illustrates the impact of incorporating different modules by examining the feature maps. Adding the dual-domain transformer notably increases the model's focus on the "car" category and regions previously ignored in the spatial domain. Furthermore, using the boundary-guided training strategy and integrating BAM greatly enhance boundary detection, as evidenced in the explicit object boundaries in the first three rows. The subsequent inclusion of the SFM enhances the model's attention to the "impervious surface," "tree," and "low vegetation" categories, aligning with the findings in Table IV, where the dual-domain transformer demonstrates superior mIoU performance in these categories.

7) Ablation Experiments of Hyperparameter: As outlined in Section IV-B, we construct the loss function by combining segmentation loss L_{seg} and boundary loss L_{bound} to optimize model training. Specifically, the hyperparameter α adjusts the weight of the boundary loss. To determine its efficacy,

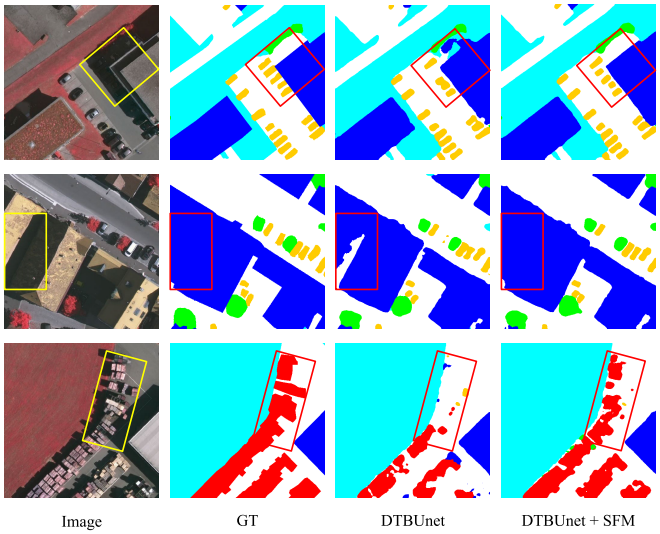


Fig. 16. Comparative visualization of segmentation outcomes from SFM ablation studies.

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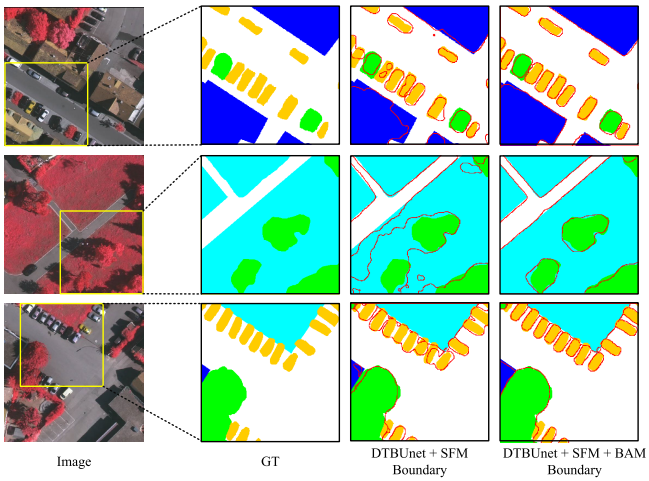


Fig. 17. Comparative visualization of segmentation results from boundary-guided training strategy ablation studies.

TABLE VIII
ABLATION EXPERIMENTS OF HYPERPARAMETER

Hyperparameter	Avg.F1 (%)	mIoU (%)	OA (%)
α = 0.3	90.1	82.3	90.8
α = 0.4	90.0	82.1	90.5
α = 0.5	**90.4**	**82.7**	**91.4**
α = 0.6	89.7	81.9	90.3
α = 0.7	89.5	81.8	90.5

we conducted ablation experiments with alpha values of 0.3, 0.4, 0.5, 0.6, and 0.7 during model training. As shown in Table VIII, the best performance of the model is achieved when alpha is taken at 0.5, with Avg.F1 reaching 90.4%, mIoU reaching 82.7%, and OA reaching 91.4%.

V. CONCLUSION

In this article, we propose a dual-domain transformer, a novel spatial and frequency-based approach for enhancing the semantic segmentation in urban RS images. Compared with the existing spatial-domain methods, concentrating solely on local and global information, our dual-domain transformer integrates FFT frequency-domain analysis for extracting RS image information to supplement the spatial-domain methods. It can capture more comprehensive features through three branches, including the global-spatial branch, the local-spatial branch, and the Fourier frequency branch. Moreover, we employ a boundary-guided training strategy to assist segmentation, facilitating precise object boundary localization and classification within intricate urban scenes. On the challenging datasets ISPRS Vaihingen, Potsdam, and Zeebrugge, our method surpasses the SOTA methods in handling the semantic segmentation task of RS urban scenes. In future work, we will extend the frequency domain characterization method to more RS image tasks.

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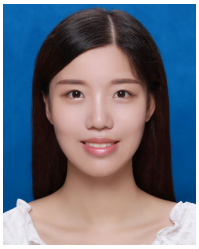
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